1.1 SYSTEMS OF LINEAR EQUATIONS

A **linear equation** in the variables $x_1, \dots, x_n$ is an equation that can be written in the form

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = b \tag{1}$$

where b and the **coefficients** $a_1, \dots, a_n$ are real or complex numbers, usually known in advance. The subscript n may be any positive integer. In textbook examples and exercises, n is normally between 2 and 5. In real-life problems, n might be 50 or 5000, or even larger.

The equations

$$4x_1 - 5x_2 + 2 = x_1 \quad \text{and} \quad x_2 = 2(\sqrt{6} - x_1) + x_3$$

are both linear because they can be rearranged algebraically as in equation (1):

$$3x_1 - 5x_2 = -2 \quad \text{and} \quad 2x_1 + x_2 - x_3 = 2\sqrt{6}$$

The equations

$$4x_1 - 5x_2 = x_1x_2 \quad \text{and} \quad x_2 = 2\sqrt{x_1} - 6$$

A **system of linear equations** (or a **linear system**) is a collection of one or more linear equations involving the same variables—say, $x_1, \dots, x_n$. An example is

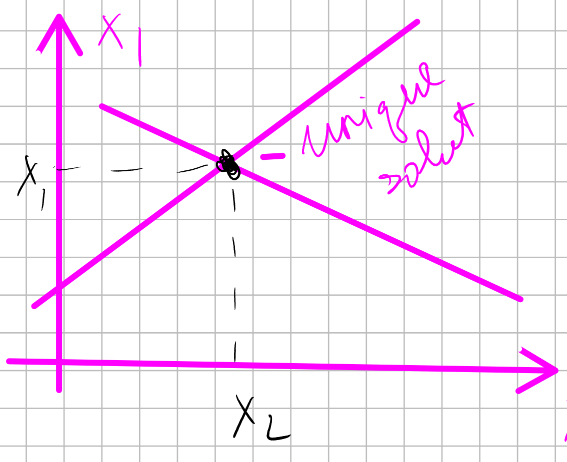
$$\begin{cases} 2x_1 - x_2 + 1.5x_3 = 8 \\ x_1 - 4x_3 = -7 \end{cases} \tag{2}$$

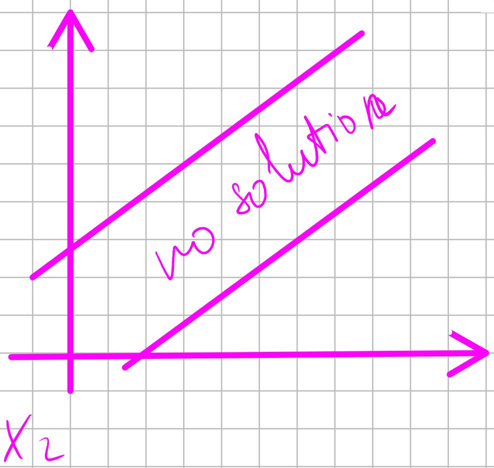
$2x + 3y = 5$ — eq of a line $\quad y = \frac{5}{3} - \frac{2}{3}x$

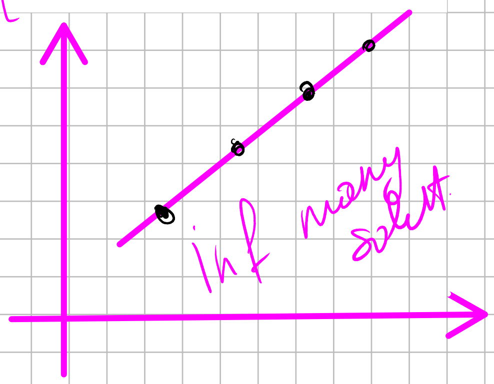
A **solution** of the system is a list $(s_1, s_2, \dots, s_n)$ of numbers that makes each equation a true statement when the values $s_1, \dots, s_n$ are substituted for $x_1, \dots, x_n$, respectively. For instance, $(5, 6.5, 3)$ is a solution of system (2) because, when these values are substituted in (2) for x_1, x_2, x_3, respectively, the equations simplify to $8 = 8$ and $-7 = -7$.

The set of all possible solutions is called the **solution set** of the linear system. Two linear systems are called **equivalent** if they have the same solution set. That is, each solution of the first system is a solution of the second system, and each solution of the second system is a solution of the first.

$$\begin{cases} x_1 - 2x_2 = -1 \\ -x_1 + 3x_2 = 3 \end{cases} \qquad \begin{cases} x_1 - 2x_2 = -1 \\ -x_1 + 2x_2 = 3 \end{cases} \qquad \begin{cases} x_1 - 2x_2 = -1 \\ \cancel{-}x_1 \cancel{+} 2x_2 = -1 \end{cases}$$







A system of linear equations has

1. no solution, or
2. exactly one solution, or
3. infinitely many solutions.

A system of linear equations is said to be **consistent** if it has either one solution or infinitely many solutions; a system is **inconsistent** if it has no solution.

Matrix Notation

The essential information of a linear system can be recorded compactly in a rectangular array called a **matrix**. Given the system

$$\left\{\begin{aligned} x_1 - 2x_2 + x_3 &= 0 \\ 2x_2 - 8x_3 &= 8 \\ 5x_1 - 5x_3 &= 10 \end{aligned}\right. \tag{3}$$

with the coefficients of each variable aligned in columns, the matrix

coefficient matrix (or **matrix of coefficients**)

$$\begin{bmatrix} 1 & -2 & 1 \\ 0 & 2 & -8 \\ 5 & 0 & -5 \end{bmatrix}$$

augmented matrix

$$\left[\begin{array}{ccc|c} 1 & -2 & 1 & 0 \\ 0 & 2 & -8 & 8 \\ 5 & 0 & -5 & 10 \end{array}\right]$$

Solving a Linear System

ELEMENTARY ROW OPERATIONS

1. (Replacement) Replace one row by the sum of itself and a multiple of another row.[1] $R_i \leftrightarrow k \cdot R_j + \ell R_m$
2. (Interchange) Interchange two rows. $R_i \leftrightarrow R_j$
3. (Scaling) Multiply all entries in a row by a nonzero constant. $k \cdot R_i$

$$\begin{cases} x_1 - 2x_2 + x_3 = 0 \\ 2x_2 - 8x_3 = 8 \\ 5x_1 - 5x_3 = 10 \end{cases}$$

pivot

$$\left[\begin{array}{ccc|c} 1 & -2 & 1 & 0 \\ 0 & 2 & -8 & 8 \\ 5 & 0 & -5 & 10 \end{array}\right] \begin{array}{c} \\ \\ R_3 - 5 \cdot R_1 \end{array} \sim \left[\begin{array}{ccc|c} 1 & -2 & 1 & 0 \\ 0 & 2 & -8 & 8 \\ 5-5\cdot 1 & 0-5\cdot(-2) & -5-5\cdot 1 & 10-5\cdot 0 \end{array}\right] \sim$$

pivot

$$\sim \left[\begin{array}{ccc|c} 1 & -2 & 1 & 0 \\ 0 & 2 & -8 & 8 \\ 0 & 10 & -10 & 10 \end{array}\right] \begin{array}{c} \\ R_2 / 2 \\ R_3 / 10 \end{array} \left[\begin{array}{ccc|c} 1 & -2 & 1 & 0 \\ 0 & \boxed{1} & -4 & 4 \\ 0 & 1 & -1 & 1 \end{array}\right] \begin{array}{c} R_1 + 2 \cdot R_2 \\ \sim \\ R_3 - R_2 \end{array}$$

$$\sim \left[\begin{array}{ccc|c} 1 & 0 & -7 & 8 \\ 0 & 1 & -4 & 4 \\ 0 & 0 & 3 & -3 \end{array}\right] \begin{array}{c} \\ \\ R_3 / 3 \end{array} \sim \left[\begin{array}{ccc|c} 1 & 0 & -7 & 8 \\ 0 & 1 & -4 & 4 \\ 0 & 0 & 1 & -1 \end{array}\right] \begin{array}{c} R_1 + 7 \cdot R_3 \\ \sim \\ R_2 + 4 \cdot R_3 \end{array}$$

pivot

$$\sim \left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 \end{array}\right] \approx \begin{cases} x_1 = 1 \\ x_2 = 0 \\ x_3 = -1 \end{cases}$$

Ans:

$(1; 0; -1)$

A rectangular matrix is in **echelon form** (or **row echelon form**) if it has the following three properties: EF

1. All nonzero rows are above any rows of all zeros.
2. Each leading entry of a row is in a column to the right of the leading entry of the row above it.
3. All entries in a column below a leading entry are zeros.

$$\begin{bmatrix} 1 & 0 & 5 & 5 \\ 0 & 2 & 0 & 5 \\ 1 & 0 & 0 & 0 \end{bmatrix} \quad 2$$

EXAMPLE 1 The following matrices are in echelon form. The leading entries (■) may have any nonzero value; the starred entries (∗) may have any value (including zero).

$$\begin{bmatrix} \blacksquare & * & * & * \\ 0 & \blacksquare & * & * \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 0 & \blacksquare & * & * & * & * & * & * & * & * \\ 0 & 0 & 0 & \blacksquare & * & * & * & * & * & * \\ 0 & 0 & 0 & 0 & \blacksquare & * & * & * & * & * \\ 0 & 0 & 0 & 0 & 0 & \blacksquare & * & * & * & * \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \blacksquare & * \end{bmatrix}$$

If a matrix in echelon form satisfies the following additional conditions, then it is in **reduced echelon form** (or **reduced row echelon form**):

4. The leading entry in each nonzero row is 1.
5. Each leading 1 is the only nonzero entry in its column.

The following matrices are in reduced echelon form because the leading entries are 1's, and there are 0's below *and above* each leading 1.

pivot column   pivot

$$\begin{bmatrix} 1 & 0 & * & * \\ 0 & 1 & * & * \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 0 & 1 & * & 0 & 0 & 0 & * & * & 0 & * \\ 0 & 0 & 0 & 1 & 0 & 0 & * & * & 0 & * \\ 0 & 0 & 0 & 0 & 1 & 0 & * & * & 0 & * \\ 0 & 0 & 0 & 0 & 0 & 1 & * & * & 0 & * \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & * \end{bmatrix}$$

Apply elementary row operations to transform the following matrix first into ~~echelon form~~ and then into reduced echelon form:

$$\begin{bmatrix} 0 & 3 & -6 & 6 & 4 & -5 \\ 3 & -7 & 8 & -5 & 8 & 9 \\ 3 & -9 & 12 & -9 & 6 & 15 \end{bmatrix}$$

$$\begin{bmatrix} 3 & -7 & 8 & -5 & 8 & 9 \\ 0 & 3 & -6 & 6 & 4 & -5 \\ 3 & -9 & 12 & -9 & 6 & \cancel{15} \end{bmatrix}_{3} \quad \begin{bmatrix} 1 & -3 & 4 & -3 & 2 & 5 \\ 0 & 3 & -6 & 6 & 4 & -5 \\ 0 & \frac{2}{2} & -4 & 4 & 2 & -6 \end{bmatrix} \quad \begin{bmatrix} 1 & -3 & 4 & -3 & 2 & 5 \\ 0 & 1 & -2 & 2 & 1 & -3 \\ 0 & 3 & -6 & 6 & 4 & -5 \end{bmatrix} \begin{matrix} +3R_2 \\ \\ -3R_3 \end{matrix}$$

$$\begin{array}{c} -5R_3 \\ -R_3 \\ \\ \end{array}\begin{bmatrix} 1 & 0 & -2 & 3 & 5 & -4 \\ 0 & 1 & -2 & 2 & 1 & -3 \\ 0 & 0 & 0 & 0 & 1 & 4 \end{bmatrix} \begin{bmatrix} 1 & 0 & -2 & 3 & 0 & -28 \\ 0 & 1 & -2 & 2 & 0 & -7 \\ 0 & 0 & 0 & 0 & 1 & 4 \end{bmatrix} \left.\begin{matrix} 1 \\ 1 \\ 1 \end{matrix}\right] \text{extra}$$

$$\begin{cases} x_1 - 2x_3 + 3x_4 - 28x_6 = 1 \\ x_2 - 2x_3 + 2x_4 - 7x_6 = 1 \\ x_5 + 4x_6 = 1 \end{cases} \qquad \begin{cases} x_1 = 1 + 2x_3 \cancel{+} 3x_4 + 28x_4 \\ x_2 = 1 + 2x_3 - 2x_4 + 7x_6 \\ x_5 = 1 - 4x_6 \end{cases}$$

pivot = basic (x_1, x_2, x_5)

others = free variables (x_3, x_4, x_6)

Find the general solution of the linear system whose augmented matrix has been reduced to

$$\left[\begin{array}{ccccc|c} 1 & 6 & 2 & -5 & -2 & -4 \\ 0 & 0 & 2 & -8 & -1 & 3 \\ 0 & 0 & 0 & 0 & 1 & 7 \end{array}\right]$$

eliminate

make it 1

$$\left[\begin{array}{ccccc|c} 1 & 6 & 2 & -5 & -2 & -4 \\ 0 & 0 & 2 & -8 & -1 & 3 \\ 0 & 0 & 0 & 0 & 1 & 7 \end{array}\right] \begin{array}{l} R_1 - R_2 \\ R_2/2 \end{array} \left[\begin{array}{ccccc|c} 1 & 6 & 0 & 3 & -1 & -7 \\ 0 & 0 & 1 & -4 & -0{,}5 & 1{,}5 \\ 0 & 0 & 0 & 0 & 1 & 7 \end{array}\right] \begin{array}{l} R_1 + R_3 \\ R_2 + 0{,}5 R_3 \end{array}$$

$$\left[\begin{array}{ccccc|c} 1 & 6 & 0 & 3 & 0 & 0 \\ 0 & 0 & 1 & -4 & 0 & 5 \\ 0 & 0 & 0 & 0 & 1 & 7 \end{array}\right] \rightarrow \text{RREF}$$

x_1, x_3, x_5 - basic var.

x_2, x_4 - free var

$$\begin{cases} x_1 + 6x_2 + 3x_4 = 0 \\ x_3 - 4x_4 = 5 \\ x_5 = 7 \end{cases} \qquad \begin{cases} x_1 = -6x_2 - 3x_4 \\ x_3 = 5 + 4x_4 \\ x_5 = 7 \end{cases}$$

general solution

Parametric solution

$$\bar{X} : \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} -6s - 3t \\ s \\ 5 + 4 \cdot t \\ t \\ 7 \end{bmatrix} = \boxed{\begin{bmatrix} 0 \\ 0 \\ 5 \\ 0 \\ 7 \end{bmatrix} + s \begin{bmatrix} -6 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} -3 \\ 0 \\ 4 \\ 1 \\ 0 \end{bmatrix}}$$

$[0 \dots 0\ 1 \mid b]$ uniq
$[0 \dots 0 \mid b]$ no solut.
$[0 \dots 0 \mid 0]$ inf. many sol

Existence and Uniqueness Questions

Existence and Uniqueness Theorem

A linear system is consistent if and only if the rightmost column of the augmented matrix is *not* a pivot column—that is, if and only if an echelon form of the augmented matrix has *no* row of the form

$$\boxed{\begin{bmatrix} 0 & \cdots & 0 & b \end{bmatrix}} \qquad \text{with } b \text{ nonzero}$$

If a linear system is consistent, then the solution set contains either (i) a unique solution, when there are no free variables, or (ii) infinitely many solutions, when there is at least one free variable.

The following procedure outlines how to find and describe all solutions of a linear system.

USING ROW REDUCTION TO SOLVE A LINEAR SYSTEM

1. Write the augmented matrix of the system.
2. Use the row reduction algorithm to obtain an equivalent augmented matrix in echelon form. Decide whether the system is consistent. If there is no solution, stop; otherwise, go to the next step.
3. Continue row reduction to obtain the reduced echelon form.
4. Write the system of equations corresponding to the matrix obtained in step 3.
5. Rewrite each nonzero equation from step 4 so that its one basic variable is expressed in terms of any free variables appearing in the equation.

$$\begin{cases} x_1 + x_2 - x_3 = 2 \\ 2x_1 - 5x_2 + x_3 = -3 \\ 3x_1 - 4x_2 = -2 \end{cases} \qquad \begin{cases} x_1 + x_2 - x_3 = 2 \\ 2x_1 - 5x_2 + x_3 = -3 \\ x_1 - 4x_2 + 4x_3 = -3 \end{cases} \qquad \begin{cases} x_1 + x_2 - x_3 = 2 \\ 2x_1 - 5x_2 + x_3 = -3 \\ 3x_3 - 4x_2 = -1 \end{cases}$$

I II III

I $\Rightarrow$

$$\left[\begin{array}{ccc|c} 1 & 1 & -1 & 2 \\ 2 & -5 & 1 & -3 \\ 3 & -4 & 0 & -2 \end{array}\right] \begin{array}{l} \\ R_2 - 2R_1 \\ R_3 - 3R_1 \end{array} \left[\begin{array}{ccc|c} 1 & 1 & -1 & 2 \\ 0 & -7 & 3 & -7 \\ 0 & -7 & 3 & -8 \end{array}\right] \begin{array}{l} \\ \\ R_3 - R_2 \end{array} \left[\begin{array}{ccc|c} & & & \\ & & & \\ 0 & 0 & 0 & -1 \end{array}\right]$$

no need to solve the rest

The last row has form $[0\ 0\ 0 \mid -1]$ — no solution

(II) $\Rightarrow$

$$\left[\begin{array}{ccc|c} 1 & 1 & -1 & 2 \\ 2 & -5 & 1 & -3 \\ 1 & -4 & 4 & -3 \end{array}\right] \xrightarrow[R_3 - R_1]{R_2 - 2R_1} \left[\begin{array}{ccc|c} 1 & 1 & -1 & 2 \\ 0 & -7 & 3 & -7 \\ 0 & -5 & 5 & -5 \end{array}\right] \xrightarrow{R_2 \leftrightarrow R_3/5}$$

$$\left[\begin{array}{ccc|c} 1 & 1 & -1 & 2 \\ 0 & 1 & -1 & 1 \\ 0 & -7 & 3 & -7 \end{array}\right] \xrightarrow[R_3 + 7R_2]{R_1 - R_2} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & -1 & 1 \\ 0 & 0 & -4 & 0 \end{array}\right] \xrightarrow{R_3/-4} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & -1 & 1 \\ 0 & 0 & 1 & 0 \end{array}\right] \xrightarrow{R_2 + R_3}$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ \boxed{0 \quad 0 \quad 1} & & & \boxed{0} \end{array}\right] \qquad \begin{aligned} x_1 &= 1 \\ x_2 &= 1 \\ x_3 &= 0 \end{aligned}$$

The last row has form $\left[\begin{array}{ccc|c} 0 & 0 & 1 & \overset{b}{0} \end{array}\right]$ – unique solution

1.4 THE MATRIX EQUATION $A\mathbf{x} = \mathbf{b}$

A fundamental idea in linear algebra is to view a linear combination of vectors as the product of a matrix and a vector. The following definition permits us to rephrase some of the concepts of Section 1.3 in new ways.

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & b_1 \\ a_{21} & a_{22} & a_{23} & b_2 \\ a_{31} & a_{32} & a_{33} & b_3 \end{bmatrix}$$

$$\downarrow \quad \downarrow \quad \downarrow \quad \downarrow$$

$$\bar{a}_1 \quad \bar{a}_2 \quad \bar{a}_3 \quad \bar{b}$$

DEFINITION

If A is an $m \times n$ matrix, with columns $\mathbf{a}_1, \ldots, \mathbf{a}_n$, and if $\mathbf{x}$ is in $\mathbb{R}^n$, then the **product of A and $\mathbf{x}$**, denoted by $A\mathbf{x}$, is **the linear combination of the columns of A using the corresponding entries in $\mathbf{x}$ as weights**; that is,

$$A\mathbf{x} = \begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_n \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n$$

Note that $A\mathbf{x}$ is defined only if the number of columns of A equals the number of entries in $\mathbf{x}$.

$A \cdot \bar{x} = \bar{b}$

$$\bar{a}_1 \cdot x_1 + \bar{a}_2 \cdot x_2 + \bar{a}_3 \cdot x_3 = \bar{b}$$

$$\begin{bmatrix} a_{11} \\ a_{21} \\ a_{31} \end{bmatrix} \cdot x_1 + \begin{bmatrix} a_{12} \\ a_{22} \\ a_{32} \end{bmatrix} \cdot x_2 + \begin{bmatrix} a_{13} \\ a_{23} \\ a_{33} \end{bmatrix} \cdot x_3 = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

$\bar{b}$ is a linear combination of the columns of the A

$$\bar{a}_1 \cdot x_1 + \bar{a}_2 \cdot x_2 + \bar{a}_3 \cdot x_3 = \bar{b}$$

Homogeneous systems $\quad A\bar{x} = \bar{b} \to 0$

$$\{A x = 0 \qquad \begin{cases} a_{11} x_1 + a_{12} x_2 = 0 \\ a_{21} x_1 + a_{22} x_2 = 0 \end{cases}$$

homog. sys of 2 var

H.S. is always consistent — it has the trivial solution

Trivial solution is the sol of all zeros